

Collision avoidance

Requirements:

- Hardware

Ultrasonic sensor, microcontroller (Arduino) , Dc motors

- Software

programming language (C)

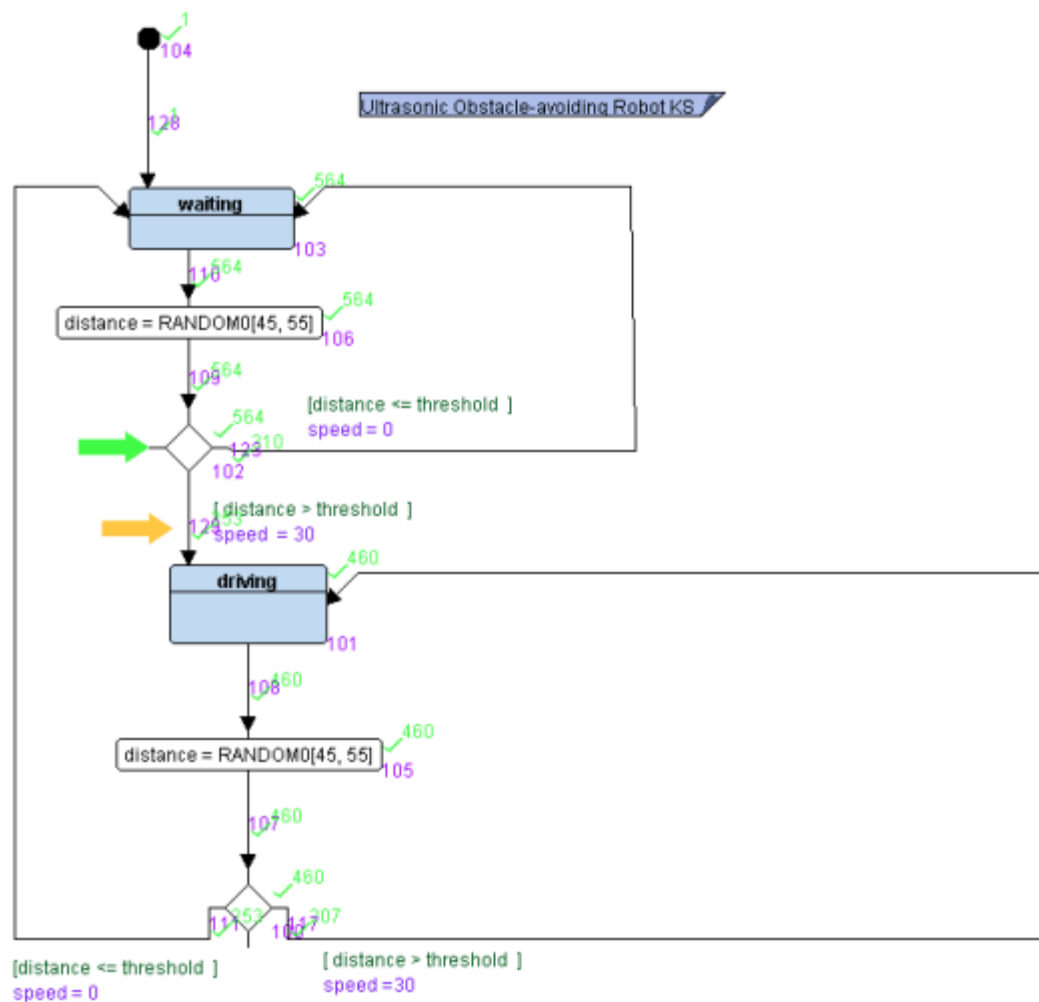
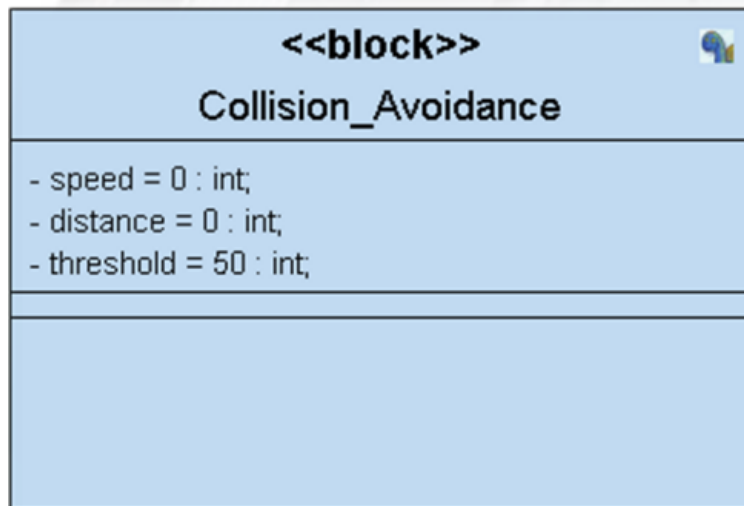
- Functional requirements

Calculate distance.

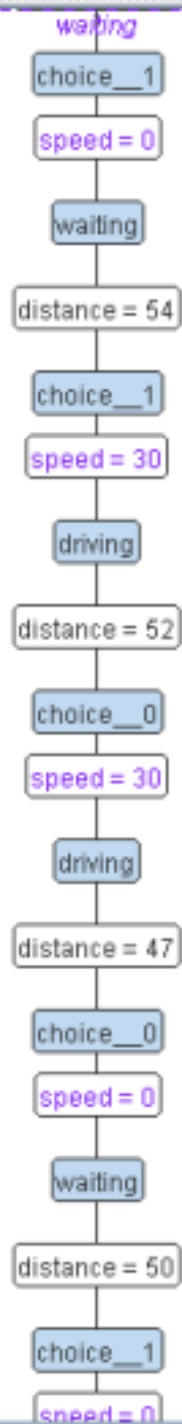
Case study:

1. The system checks if there is no obstacle in 50cm in front of the car, the car moves at 30cm/s.
2. If there is an obstacle in 50cm in front of the car the car stops.

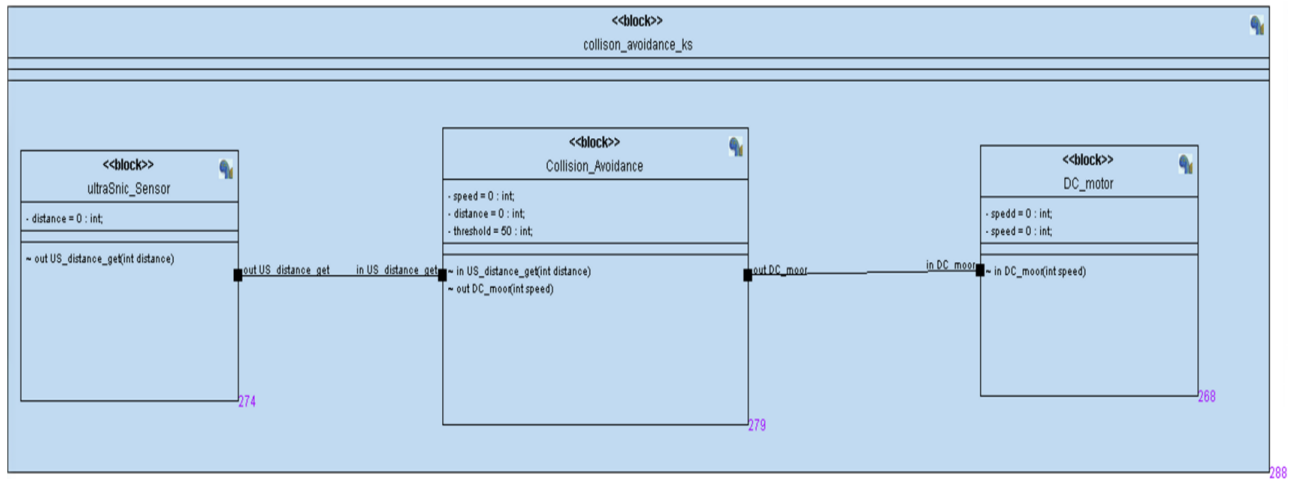
- Design diagrams for one module system.



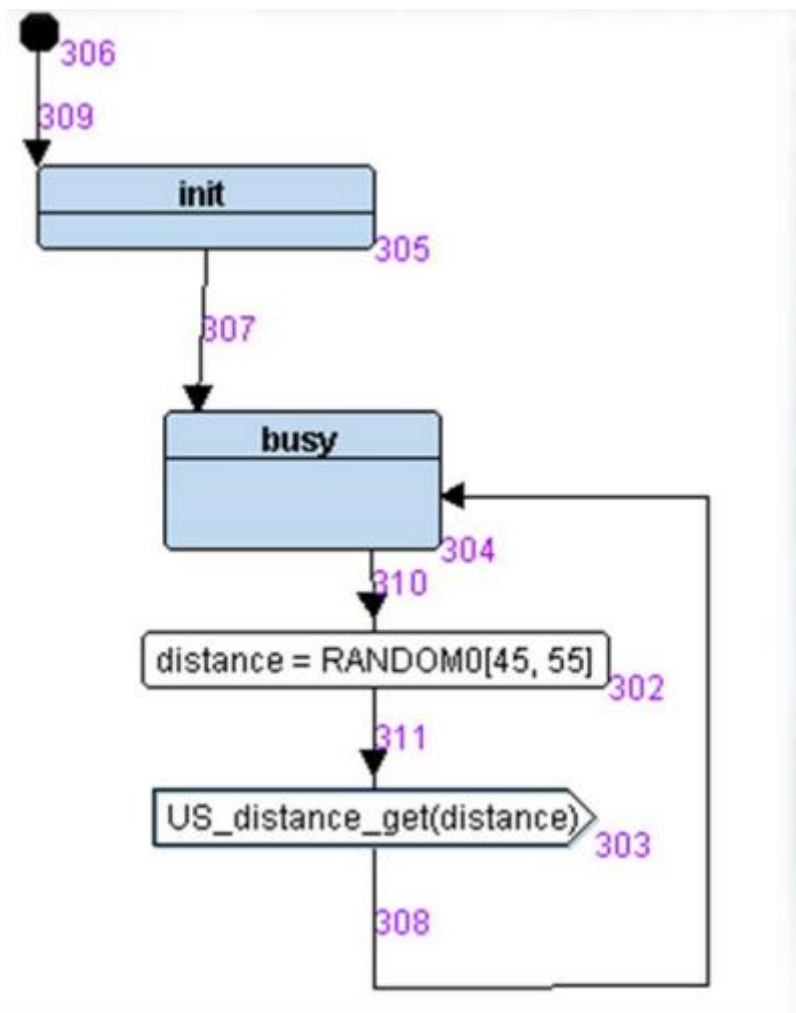
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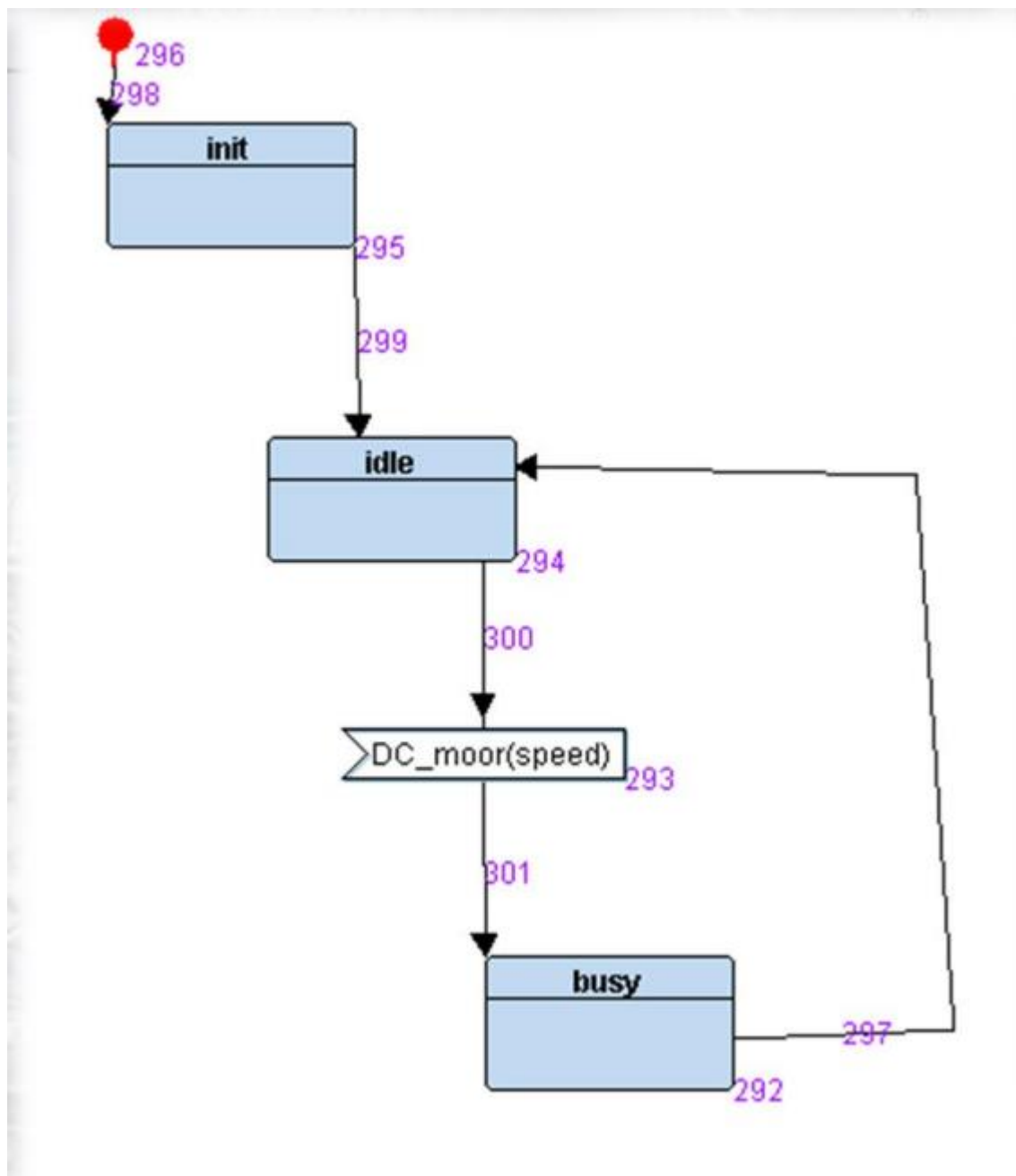
- Design diagrams for multiple modules system.



1. Ultrasonic module



2. DC motor module



3. CA module

